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The Effects of Aerobic Exercise on Cognitive Function of Alzheimer's Disease Patients

Si-Yu Yang[1,§], Chun-Lei Shan[2a,2b,§], He Qing[3], Wei Wang[1], Yi Zhu[2], Meng-Mei Yin[1], Sergio Machado[4], Ti-Fei Yuan[*,5] and Ting Wu[*,1]

Ti-Fei Yuan

[1]*Department of Neurology, The First Affiliated Hospital of Nanjing Medical University, Nanjing 210029, China*

[2a]*Shanghai University of Traditional Chinese Medicine, Shanghai, China*

[2b]*Department of rehabilitation medicine, First affiliated hospital of Nanjing Medical University, Nanjing, China*

[3]*Department of Neurology, The First Hospital of Xuzhou, Xuzhou, 221002, Jiangsu, China*

[4]*Laboratory of Panic and Respiration, Institute of Psichiatry, Federal University of Rio de Janeiro (IPUB/UFRJ), Rio de Janeiro, Brazil; Physical Activity Neuroscience Laboratory, Physical Activity Sciences Postgraduate Program - Salgado de Oliveira University, Niterói, Brazil*

[5]*School of Psychology, Nanjing Normal University, Nanjing 210097, China*

Ting Wu

Abstract: To evaluate the effect of moderate intensity of aerobic exercise on elderly people with mild Alzheimer's disease, we recruited fifty volunteers aged 50 years to 80 years with cognitive impairment. They were randomized into two groups: aerobic group (n=25) or control group (n=25). The aerobic group was treated with cycling training at 70% of maximal intensity for 40 min/d, 3 d/wk for 3 months. The control group was only treated with heath education. Both groups were received cognitive evaluation, laboratory examination before and after 3 months. The results showed that the Minimum Mental State Examination score, Quality of Life Alzheimer's Disease score and the plasma Apo-a1 level was significantly increased ($P<0.05$), the Alzheimer's Disease Assessment Scale-cognition score, Neuropsychiatric Inventory Questionnaire score was significantly decreased ($P<0.05$) in aerobic group before and after 3 months in aerobic group. For the control group, there was no significant difference in scores of Alzheimer's Disease Assessment Scale-cognition, Neuropsychiatric Inventory Questionnaire, Quality of Life Alzheimer's Disease, Apo-a1 ($P>0.05$), while Minimum Mental State Examination scores decreased significantly after 3 months ($P<0.05$). In conclusion, moderate intensity of aerobic exercise can improve cognitive function in patients with mild Alzheimer's disease.

Keywords: Alzheimer's disease, aerobic exercise, cognitive function, moderate intensity.

INTRODUCTION

Alzheimer's disease (AD) is a progressive neurodegenerative disease, the most common cause of dementia [1]. AD is characterized by global cognitive decline involving memory, judgment, orientation, and reasoning. Senile plaques (SP) and neurofibrillary tangles (NFTs) are important pathological features of AD [2]. Population epidemiologic studies have revealed the incidence of AD doubles every 5 years after the age of 65 [3]. The course of AD averages 3-9 years, patients often die because of various complications in a completely helpless state [4]. Moreover, AD will bring enormous emotional and financial burden on patients, their families, and society [5]. However, there is no effective cure or preventative therapy yet [6]. So we have an urge to find an innovative approach in AD treatment. It is well known that aerobic exercise is beneficial to physical and mental disorder, such as cardiovascular disease, cancer, obesity, depression and anxiety.

Recent evidences have shown the effectiveness of aerobic exercise to prevent or delay the development of AD [7]. The protective mechanisms of aerobic exercise on AD remain unknown [8]. Also, the effect of aerobic exercise on patients with AD remains controversial [9, 10]. Foreign clinical research [11, 12] found that AD patients with small intensity exercise such as walking, help to improve the symptoms of AD patients. However, the effect of high intensity aerobic exercise on the patients of AD is still insufficient. Frederiksen [13] performed a small sample of patients with middle-high intensity aerobic exercise intervention in the clinical trials of patients with AD, the results showed that the patients with AD had good tolerability and safety of middle-high intensity aerobic

*Address correspondence to these authors at the (Ting Wu) Department of Neurology, The First Affiliated Hospital of Nanjing Medical University, China; Tel: +86 18260027656; Fax: +86 02568136050;
E-mail: wuting80000@gmail.com and (Ti-Fei Yuan) School of Psychology, Nanjing Normal University, China; Tel/Fax: +86 25 8359 8516;
Email: ytf0707@126.com

§Si-Yu Yang and Chun-Lei Shan contributed equally to this study.

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exercise. In China, there is still a lack of theoretical research and clinical practice for the exercise therapy of AD. To the majority of AD patients' family caregivers, the pressure is great. It's also a heavy burden to the hospital because of the long treatment cycle and low ward bed turnover rate.

In this study, we are going to investigate whether moderate-intensity aerobic exercise can improve cognitive function of patients with AD and provide a reliable basis for the interventions of aerobic exercise in AD treatment. Through aerobic exercise, a new non-drug intervention, to slow the progression of cognitive impairment in patients with AD and explore a concrete and effective aerobic exercise program, in order to further promote it to the community and provides an effective non-pharmacological treatment options to the AD patients rehabilitative treatment.

MATERIALS AND METHODS

Subjects

Participants aged 65 to 80 were recruited from the neurology clinic of the first people's hospital in Jiangsu province from March 2012 to May 2013. They were all in stable condition and in accordance with the latest AD diagnostic criteria; Minimum Mental State Examination (MMSE)≤24 and ≥10. Subjects who were diagnosed with vascular dementia according to the NINDS-AIREN, Modified Hachinski ischemic scale (MHIS)>4 and cannot be coordinate with cognitive function tests (including blindness, deafness, severe language disorder and so on) were excluded. This study was approved by the local ethics review board, and informed consent was obtained from volunteers/ guardians (Fig. **1**).

Intervention Methods

Exercise Group: Cycling training was performed in the rehabilitation clinic of Jiangsu Province Hospital. Exercise intensity is moderate, 70% maximum heart rate (MHR). The maximum heart rate indicates the reserve of maximum movement capacity, and is commonly used to determine the intensity of aerobic exercise. The calculation formula is 220 minus the actual age. Each training lasted for 40min (5min warm-up, 30min target intensity exercise, 5min reorganization movement). The training time of the initial stage was 25-30min, and the exercise load was 0.5kg·M. Then according to patient tolerance gradually increased training time and training intensity. One week later, the training time was increased to 40min (mainly increased the strength of the target intense exercise time, warm-up and reorganization movement were unchanged), cycling load was increased to 1.0kg·M (to reach the target heart rate) and subsequently maintained the training intensity to the end. The training progress was supervised by therapists.

Control Group: The subjects were under health education for 3 months. We distributed AD related information to the control group patients, asking the patients to live by the past, not to participate in the aerobic exercise. Every two weeks we called the control group patients to hold a lecture and asked about their recent situation.

Cognitive Evaluation

(1) MMSE (Minimum Mental State Examination): *a widely used inspection tool in AD patients, assessment of time, location, orientation, word register, memory, attention, force calculation, language and visual capacity. The total score is 30*

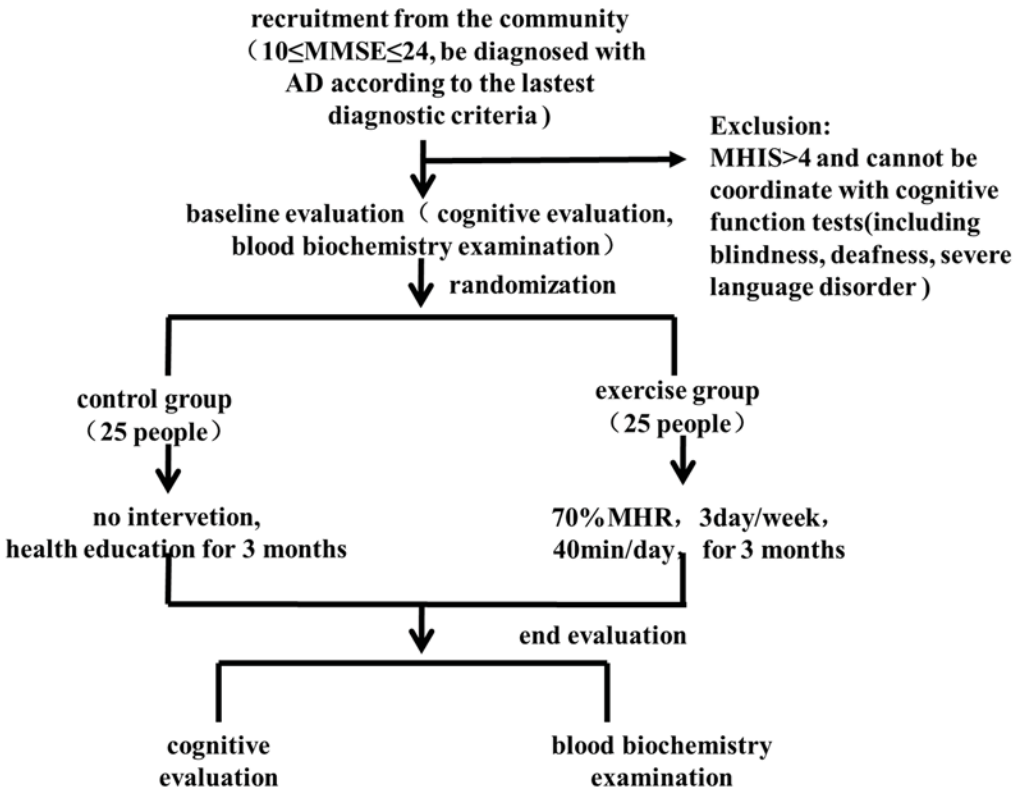

Fig. (1). Experiment procedure.

points. The lower the score, the worse the cognitive level is. In our study, MMSE was used mainly for the assessment of pre-screening and cognitive function.

(2) Adas-cog (Alzheimer 's Disease Assessment Scale-cognition): *This scale has been widely recognized for the evaluation of the severity of AD. It consists of 12 items, verbal memory, naming, executive instructions, structural practice, intentionality exercises, directional force, word identification, memory test instruction, spoken language expression ability, find words, spoken language comprehension ability and attention. From memory, language, operational ability and attention of these 4 aspects to assess cognitive function. The score ranged from 0 to 75, and the higher the score is, the worse the cognitive level is.*

(3) *NPI-Q (*Neuropsychiatric Inventory Questionnaire): The scale is mainly informed from direct caregivers. *The 12 behavioral and psychological symptoms were evaluated by caregivers, including delusions, hallucinations, agitation, depression, anxiety, apathy, euphoria, disinhibition behavior, abnormal movements, nocturnal behavior disorders, eating abnormal. The product of each symptom frequency (0~4 points) and severity (0~3 points) was the score (0~12 points) and total score is 144 points. The higher score shows a more severe symptoms.*

(4) Qol-AD (Quality of Life Alzheimer's Disease): *The scale has good reliability and validity. It has a total of 13 entries using Likert five scale method. The total score of the entries was 13 to 65, the higher the score is, the better the quality of life for the patients is.*

Blood Biochemistry Examination

Each participant was taken 3ml blood in the median cubital vein after fasting overnight and automatic biochemical analyzer was used to determination of plasma lipids and cholesterol. All blood samples would be sent to the laboratory department of hospital for blood biochemical examination using automatic biochemical analyzer after centrifugation. The items of biochemical examination included plasma triglyceride, low density lipoprotein, apolipoprotein a1 and total cholesterol.

Statistical Analysis

Data was expressed as mean ± SD. All statistical analysis was performed using SPSS version 21.0 software. Gender composition was performed using corrected Chi-square test. The level of education was performed using K Independent Samples. The baseline analysis between the exercise group and control group was performed using variance analysis. The paired samples t-test was used to analyze the cognitive evaluation and blood biochemistry examination results before and after 3 month in exercise and control group, respectively. Statistical significance was accepted at $p < 0.05$.

RESULTS

Analysis on the basic condition of two groups have no statistically significant (P>0.05). The baseline characteristics of each treatment group are listed in Table **1**. Each group was comparable for all demographic and clinical characteristics (p> 0.05) which included: age, sex, education background and MMSE score (Table **1**).

Analysis on the results of aerobic group have showed that the scores of MMSE and Qol-AD were increased and Adas-cog, NPI was decreased in aerobic group after 3 months in our study. And there is a statistically significant on the MMSE, Adas-cog, NPI, Qol-AD before and after 3 months in aerobic groups (respectively, P=0.000, P=0.004, P=0.004, P=0.013). It indicated that aerobic exercise could improve AD patients' cognitive function, mental state and the quality of life. MMSE score decreased significantly in control group (P=0.0037) while ADAS-cog score, NPI score, Qol-AD score changes have no statistically significant (respectively, P=0.080, P=0.305, P=0.225) (Table **2**).

Analyses on the results of laboratory examination have indicated that Apo-a1 increased significantly (P<0.0041) while the plasma TC, TG and LDL level showed no statistically significance in aerobic group before and after 3 month (respectively P=0.888, P=0.815, P=0.086). In control group, the plasma TC, TG and LDL and Apo a1 level have no statistically significance (respectively, P=0.927, P=0.148, P=0.011, P=0.270) (Table **3**).

DISCUSSION

The scores of MMSE and Qol-AD were increased and Adas-cog, NPI was decreased in aerobic group after 3 months in our study. And there is a statistically significant on the MMSE, Adas-cog, NPI, Qol-AD before and after 3 months in aerobic groups. The score of MMSE decreased significantly in control group (P<0.05). Results indicated that the cognitive function, mental status and self-care ability of AD patients improved in aerobic group while declined in control group. It is consistent with previous studies.

Laboratory examination showed Apo-a1 increased significantly in aerobic group after 3 months (P<0.05). Several studies have showed that decreased Apo-a1 levels are associated with the severity of AD and lipid metabolism during remyelination and axonal regeneration after nerve injury. It is entirely supplied to the nerve tissue by plasma lipoproteins. Therefore, a decrease of Apo-a1 would lead to a low level in nerve tissue and constitute a risk factor for AD [14]. Studies have showed that plasma TG and LDL level of AD patients is generally higher than normal, and exercise can lower plasma levels of both [15]. In this study, the plasma TG and LDL level of aerobic group showed a slight increase after 3 months, but the result has no statistically significant. There may be two reasons for the conflicting results. Firstly, three months of training is not enough to cause the change of blood biochemical indicators. Secondly, maybe because we did not control the diet factor. We informed patients of each blood biochemical examination report, so the patients could subjectively change their eating habits according to their own examination results. In future

Table 1. Comparison of the baseline between the two groups.

Group	Case	Sex		Age	Educational Background	MMSE
		Male	Female			
Aerobic group	25	10	15	72.00±6.69	21.18	21.33±2.24
Control group	25	7	18	71.92±7.28	19.73	20.00±3.50

Table 2. Comparison of cognition function between the two groups.

Group	Case	MMSE	ADAS-cog	NPI	Qol-AD
Aerobic Group	25				
V0 observation period		21.33±2.24	17.50±6.76	11.21±8.77	29.17±4.51
V1 observation period		22.83±2.75	14.63±6.83	7.96±8.15	30.67±4.84
		0.000	0.004	0.004	0.013
Control Group	25				
V0 observation period		20.00±3.50	21.46±8.73	11.35±11.60	30.85± 4.76
V1 observation period		19.54±3.43	22.38 ±8.12	12.54 ±14.40	30.19 ±4.53
P value		0.037	0.08	0.305	0.225

Note: V0 Observation Period=Before the Recruitment; V1 Observation Period=3 Months After the Recruitment; Compared with V0 Observation Period, *P<0.05.

Table 3. Comparison of blood test between the two groups.

Group	Case	TC	TG	LDL	Apo a1
Aerobic Group	25				
V0 observation period		5.35±0.94	1.37±0.55	3.19±0.72	1.38±0.20
V1 observation period		5.38±1.06	1.52±0.68	3.22±0.85	1.49±0.34
P		0.888	0.815	0.086	0.041
Control Group	25				
V0 observation period		5.14±0.88	1.23±0.64	2.86±0.84	1.58±0.45
V1 observation period		5.15±0.92	1.43±0.76	3.06 ±0.64	1.49±0.27
P value		0.927	0.148	0.011	0.270

Note: TG=Triglyceride, LDL=Low Density Lipoprotein, Apo-a1=Apolipoprotein a1, TC= Total Cholesterol. V0 Observation Period=Before the Recruitment; V1 Observation Period=3 Months After the Recruitment; Compared with V0 Observation Period, *P<0.05.

studies, we will pay more attention to this and control possible influencing factors strictly.

In the past, there is accumulated evidence proved that exercise improved progressive cognitive declines in patients with AD [16]. Aerobic exercise is beneficial to health, which can extend life expectancy and reduce the risk of chronic diseases [17]. The mechanism of exercise for improving cardiovascular and lung function are well established, but not for cognitive function [18]. Now more and more researchers pay attention to the role of aerobic exercise on cognitive function. Research on animals has demonstrated improved cognitive function as a result of aerobic exercise [19]. From the aspect of cellular and molecular, an increase in cell survival and cell proliferation in the dentate gyrus of the hippocampus is one of the most consistently observed effects of aerobic exercise treatment [20, 21]. AD are characterized by a marked reduction in the number of neurons in the hippocampus, which may be mitigated by increased neurogenesis due to aerobic activity partly [22]. Moreover, researches proved that the beneficial effects of aerobic exercise on AD may be mainly correlative with declined oxidative stress [23], and it is possible to prevent AD by managing the "redox" state [24]. Aerobic exercise not only protects synapses from oxidative stress, but also improves plasticity of brain endothelial cells and facilitates angiogenesis by stimulating the expression of angiogenic factors, for instance, vascular endothelial growth factor and angiotensin II [25, 26].

The epidemiological studies showed that aerobic exercise played an important role in the prevention of AD. Weuve *et al.* [27] divided 18766 elderly women over 60 years old into five groups according to the intensity of exercise. The results showed that cognitive evaluation improvement of those in the high intensity exercise group were significantly better

than those in the low intensity exercise group after ten years follow-up. Baker *et al*. [28] found that high intensity exercise could improve the cognitive function of patients with mild cognitive impairment by a clinical randomized controlled study. Studies have showed that the brain hippocampus volume of AD patients who performed aerobic exercise three times a week for one year increased by 2%, which is conducive to the delayed brain function recession [29]. At the present stage, the majority of the treatment for AD is still in a single drug treatment pattern. According to the more and more basic and clinical evidence in recent years, research results showed that aerobic exercise could benefit patients with AD. AD patients accepted aerobic exercise therapy should be paid more and more attention, including aerobic exercise mode, time and intensity factors should be fully taken into account. These results also suggested that AD patients of middle and high strength with aerobic exercise has a good tolerability and safety, can improve cognitive function in AD patients, and long-term moderate intensity of aerobic exercise makes patients benefit more.

According to the results above, we can put forward a hypothesis that aerobic exercise is a protective factor for AD and play a role in AD prevention. American College of Sports Medicine and the American Heart Association suggest that older people should do moderate-intensity aerobic exercise at least 30min one day [30]. Due to the limited effect of pharmacotherapy of AD, early prevention is particular important.

In the end, this study has several limitations. The first is the small sample size. The second is the subjective of Neurological and Psychiatric Evaluation. Although we avoid subjective factors as much as possible, cannot completely exclude the effect of subjective factors in evaluation. The third is lack of an active control group. In the present study, it does not allow us to exclude the possibility that some benefits may gain from social contact rather than strictly to the physical training alone. The forth is lake of follow-up. Moreover, in this study, we just investigate the effect of one single form and medium intensity of aerobic exercise. In our further study, we will control the influence of these limitations above and discuss the intervention of different intensity and form of aerobic exercise on cognitive function.

CONCLUSION

In conclusion, the present study for aerobic exercise to improve the symptoms of AD patients provides strong evidence, but aerobic exercise intervention on AD is a complex process of a multi system, multi way, still need more basic and clinical studies to confirm the role and mechanisms, and to explore the more reasonable intervention model. We intend to further observe the exercises of different intensity and different intervention time of aerobic training on cognitive function in patients with AD, constantly optimize the exercise program, and provide a kind of effective method for the treatment of non -drug rehabilitation of patients with AD. We will further explore the suitable aerobic training method for community.

LIST OF ABBREVIATIONS

AD	=	Alzheimer's Disease
Adas-cog	=	Alzheimer 's Disease Assessment Scale-Cognition
Apo-a1	=	Apolipoprotein a1
LDL	=	Low Density Lipoprotein
MMSE	=	Minimum Mental State Examination
NPI	=	Neuropsychiatric Inventory Questionnaire
Qol-AD	=	Quality of Life Alzheimer's Disease
TG	=	Triglyceride

CONFLICT OF INTEREST

The authors confirm that this article content has no conflict of interest.

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